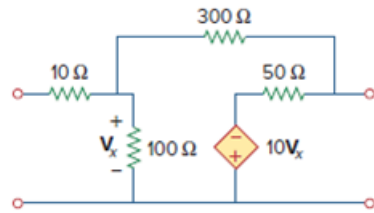


## Problem

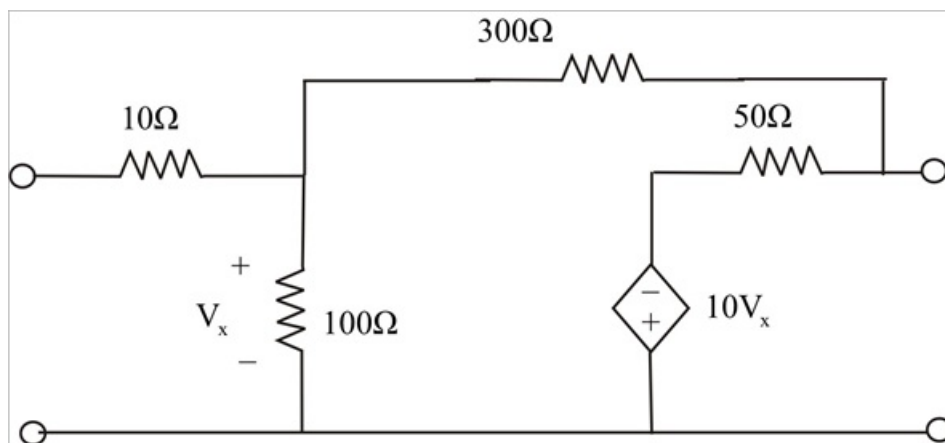
Obtain the  $h$  and  $g$  parameters of the two-port in Fig. 19.92.

Figure 19.92



## Step-by-step solution

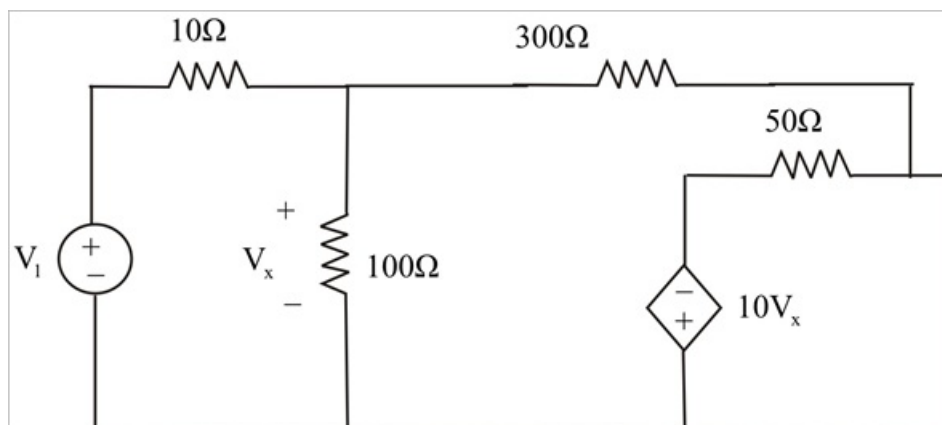
## Step 1 of 10



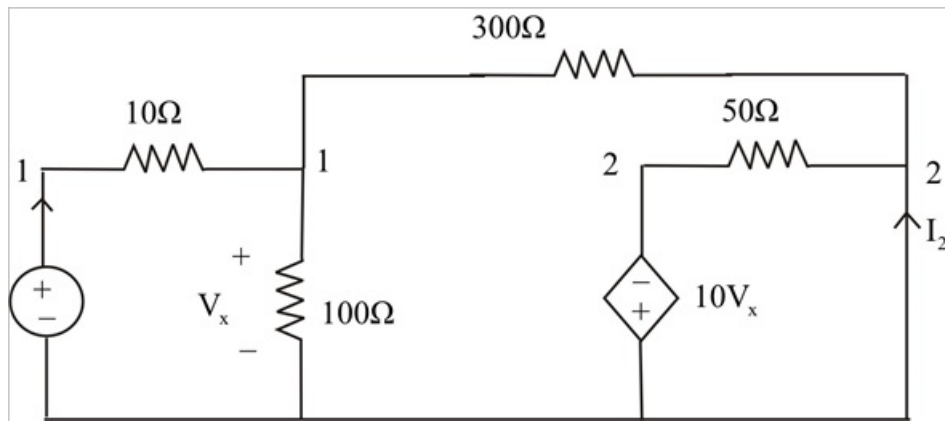
## Step 2 of 10

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0} \quad h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} \quad h_{12} = \left. \frac{V_1}{V_2} \right|_{I_2=0} \quad h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

For  $V_2 = 0$  the circuit becomes



## Step 3 of 10



**Step 4** of 10

Since 2 is connected to ground  $V_2 = 0$  at Node 1  $I_1 = \frac{V_1 - V_1'}{10} \Rightarrow 10I_1 \quad -(1)$

$$\begin{aligned} \text{At Node 1'} \quad \frac{V_1 - V_1'}{10} &= \frac{V_1}{100} + \frac{V_1' - V_2}{300} \\ \Rightarrow \frac{V_1 - V_1'}{10} - V_1' - \frac{V_1'}{30} &= 0 \\ \Rightarrow 30V_1 - 3V_1' - 30V_1' - V_1' &= 0 \\ \Rightarrow 30V_1 - 34V_1' &= 0 \Rightarrow V_1 = \frac{34}{30}V_1' \quad -(2) \end{aligned}$$

**Step 5** of 10

Substituting in (1)

$$V_1 - \frac{30V_1'}{34} = 10I_1$$

$$\frac{4}{340}V_1 = I_1 \Rightarrow \frac{V_1}{I_1} = \frac{340}{4} = 85\Omega$$

$V_1' = V_x$  substituting in equation (1) & (2) we have

$$V_1' = V_x = 10I_1 \quad -(1)$$

$$30V_1 = 34V_x = 0 \quad -(2)$$

At Node 2

$$-\frac{10V_x - V_2}{50} + \frac{V_x - V_2}{300} + I_2 = 0$$

$$I_2 = +\frac{V_x}{5} - \frac{V_x}{300} - \frac{V_2}{50} + \frac{V_2}{300}$$

$$\Rightarrow I_2 = \frac{+60V_x - V_x}{300} + \frac{-6V_2 + V_2}{300} - 3a$$

$$\Rightarrow 59V_x - 5V_2 = 300I_2 \quad -(3)$$

At Node 2'

$$\frac{10V_x - V_2}{50} = 0 \Rightarrow V_x = \frac{V_2}{10}$$

$$I_2 = \left(6V_2 - \frac{V_2}{10}\right) \times \frac{1}{300} - \frac{6V_2 - V_2}{300}$$

$$\Rightarrow 5.9V_2 - 5V_2 = 300I_2$$

$$\Rightarrow I_2 = \frac{0.9}{300}V_2$$

$$V_1 = \frac{34}{30}V_x = \frac{34}{300}V_2 \quad \text{Substituting in (1)}$$

Step 6 of 10

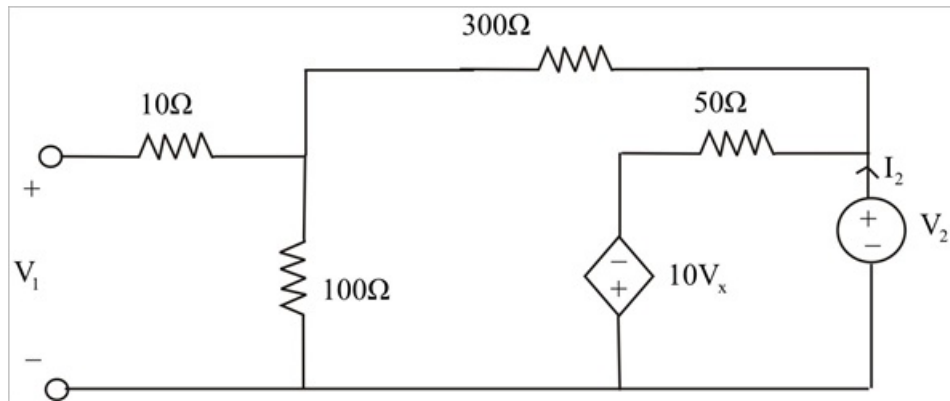
$$\frac{34}{300}V_2 - \frac{V_2}{10} = 10I_1$$

$$\left(\frac{3000}{9}\right)I_2 \left(\frac{34}{300} - \frac{1}{10}\right) = 10I_1$$

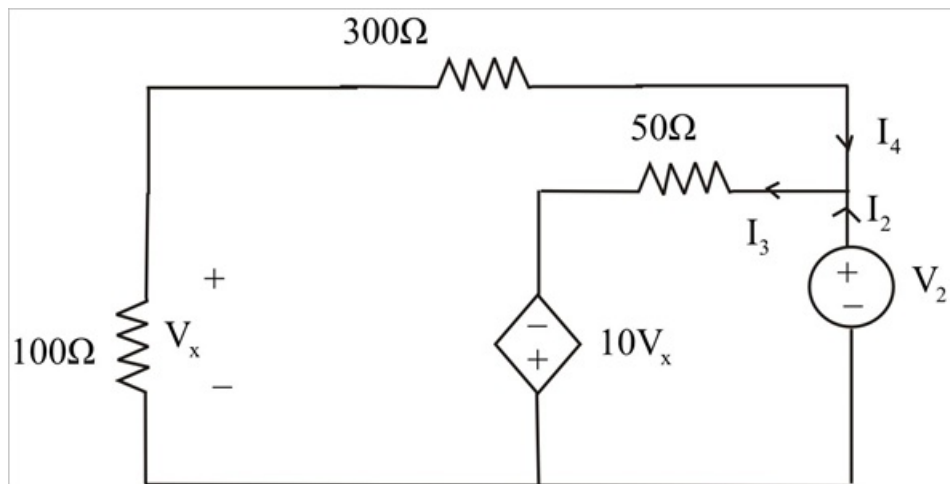
$$\frac{I_2}{I_1} = \frac{1}{\left(\frac{300}{9}\right)\left(\frac{34}{300} - 0.1\right)} = \frac{1}{0.444} = 2.254$$

Step 7 of 10

For  $I_1 = 0$



Step 8 of 10



Step 9 of 10

Current through 50Ω resistance

$$\frac{V_2 + 10V_x}{50} = I_3$$

$$I_4 = \frac{V_2}{400} A$$

$$V_1 = 100 \times I_4 = \frac{100 \times V_2}{400}$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{1}{4} = 0.25$$

At Node 2  $I_2 = I_3 + I_4$

$$\Rightarrow \frac{V_2 + 10\left(\frac{V_2}{400}\right) \times 100}{50} + \frac{V_2}{400}$$

$$\frac{I_2}{V_2} = 0.0725S$$

$$h = \begin{bmatrix} 8.5\Omega & 2.25 \\ 0.25 & 0.725S \end{bmatrix}$$

$$\Delta h = 6.1625$$

$$g = \begin{bmatrix} 0.012S & -0.04 \\ -0.365 & 13.793\Omega \end{bmatrix}$$